

TRANSFORMATION TECHNIQUES • DENNIS G. ZILL & COURSE SYLLABUS W3

Chapter 2.5: Solutions by Substitution (Bernoulli)

Transforming nonlinear differential equations into solvable linear forms using standard power substitutions.

Zill §2.5 (pp. 99–108)

Syllabus Week 3 Core

Midterm 1 Classic

PLAIN-ENGLISH INTUITION

When an equation is nonlinear, direct integration fails. But if the nonlinearity has the specific structure of a **Bernoulli equation** ($y' + Py = fy^n$), substituting $u = y^{1-n}$ miraculously linearizes the entire equation! We then solve for $u(x)$ using standard integrating factors and back-substitute to find $y(x)$.

THE BERNOULLI DIFFERENTIAL EQUATION & UNIVERSAL SHORTCUT

A differential equation of the form:

$$\frac{dy}{dx} + P(x)y = f(x)y^n \quad (n \neq 0, 1)$$

is transformed into a **pure linear ODE** for $u(x)$ by setting $u = y^{1-n}$:

$$\frac{du}{dx} + (1-n)P(x)u = (1-n)f(x)$$

Master Worked Example 1 • Bernoulli ODE (Zill Ex 2, p. 101)**MIDTERM EXAM CLASSIC**

Problem: Solve the differential equation $x \frac{dy}{dx} + y = x^2 y^2$ for $x > 0$ with $y(1) = 1$.

1 Put in Standard Form:

Divide by x : $\frac{dy}{dx} + \frac{1}{x}y = xy^2$.

Identify: $P(x) = \frac{1}{x}$, $f(x) = x$, $n = 2$.

2 Apply Substitution $u = y^{1-n}$:

$$u = y^{1-2} = y^{-1} = \frac{1}{y} \implies \frac{du}{dx} = -y^{-2} \frac{dy}{dx}.$$

Multiply standard form by $-y^{-2}$:

$$\frac{du}{dx} - \frac{1}{x}u = -x \quad (\text{Standard Linear!})$$

3 Solve the Linear Equation for $u(x)$:

$$\mu(x) = e^{\int -\frac{1}{x} dx} = e^{-\ln x} = \frac{1}{x}.$$

$$\frac{d}{dx} \left[\frac{1}{x} u \right] = \frac{1}{x} (-x) = -1.$$

$$\text{Integrate: } \frac{u}{x} = -x + C \implies u(x) = Cx - x^2.$$

4 Back-Substitute & Apply $y(1) = 1$:

$$\frac{1}{y} = Cx - x^2 \implies y(x) = \frac{1}{Cx - x^2}.$$

$$y(1) = 1 \implies 1 = \frac{1}{C-1} \implies C = 2.$$

$$\text{Solution: } y(x) = \frac{1}{2x - x^2}. \quad (\text{Lost solution: } y \equiv 0).$$

THE $(1 - N)$ DISTRIBUTION TRAP

When applying the linear equation shortcut, remember that $(1 - n)$ multiplies BOTH $P(x)$ and $f(x)$! In the problem above, $n = 2 \implies 1 - n = -1$. Neglecting the negative sign on $-x$ is the most common exam mistake.

2 Homogeneous Equations & Linear Argument Substitutions

HOMOGENEOUS EQUATIONS (DEGREE N)

A function $f(x, y)$ is homogeneous of degree n if $f(tx, ty) = t^n f(x, y)$.

If $M(x, y)$ and $N(x, y)$ have the same degree, then $\frac{dy}{dx} = F\left(\frac{y}{x}\right)$.

Substitution: $y = ux \implies dy = u dx + x du$.

Always transforms the DE into a **separable equation** in u and x !

LINEAR ARGUMENT SUBSTITUTIONS

Equations where x and y only appear inside a linear combination:

$$\frac{dy}{dx} = f(Ax + By + C)$$

Substitution: $u = Ax + By + C$.

$$\frac{du}{dx} = A + B \frac{dy}{dx} \implies \frac{dy}{dx} = \frac{1}{B} \left(\frac{du}{dx} - A \right).$$

Always transforms into a **separable equation**!

Master Worked Example 2 · Homogeneous Substitution (Zill p. 100)

STEP-BY-STEP BREAKDOWN

Problem: Solve $(x^2 + y^2) dx + (x^2 - xy) dy = 0$.

1 Verify Homogeneity:

Terms x^2, y^2, x^2, xy all have total algebraic degree 2. Homogeneous!

2 Substitute $y = ux \implies dy = u dx + x du$:

$$(x^2 + u^2 x^2) dx + (x^2 - ux^2)(u dx + x du) = 0.$$

Factor out x^2 (for $x \neq 0$):

$$(1 + u^2) dx + (1 - u)(u dx + x du) = 0.$$

$$(1 + u^2 + u - u^2) dx + x(1 - u) du = 0.$$

$$(1 + u) dx + x(1 - u) du = 0$$

3 Separate Variables:

$$\frac{dx}{x} + \frac{1-u}{1+u} du = 0 \implies \frac{dx}{x} + \left(\frac{2}{1+u} - 1 \right) du = 0.$$

4 Integrate Both Sides:

$$\ln|x| + 2 \ln|1+u| - u = c_1 \implies \ln|x(1+u)^2| - u = c_1.$$

5 Back-Substitute $u = y/x$:

$$\ln \left| x \left(1 + \frac{y}{x} \right)^2 \right| - \frac{y}{x} = c_1 \implies \ln \left| \frac{(x+y)^2}{x} \right| - \frac{y}{x} = c_1.$$

$$(x+y)^2 = Cxe^{y/x}$$

THE PRODUCT RULE OMISSION TRAP

When substituting $y = ux$, you **MUST** use the product rule for dy : $dy = u dx + x du$. Writing $dy = u dx$ is an instant disqualifier that wrecks the entire problem!